

Group5_RMarkdown_Answers

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Section 2: Formatting Text Using RMarkdown Syntax

Objective: learn how to format the appearance of your document in your chosen output format. Author: Arnav Pandey (a_pandey2@uncg.edu)

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2.1. Headings

.....

Task: Read the question below, click **Knit** to render the document and return here to answer the question.

Question: What difference do you notice about the above line that starts with `# Section 2: Formatting a Report Using RMarkdown Syntax` compared to the line `Objective: learn to format how your document appears in the output of your choice?`

Answer: The font is large and bold in line starting with `# section 2.` while the other line seems to be normal.

Placing a hashtag (#) at the beginning of a line turns it into a heading. RMarkdown supports six levels of headings.

Task: Click Knit to observe how different heading levels below appear in the rendered output.

Line 1: This is an example to show different levels of headings in RMarkdown.

Line 2: Line 1: This is an example to show different levels of headings in RMarkdown.

Line 3: This is an example to show different levels of headings in RMarkdown.

Line 4: This is an example to show different levels of headings in RMarkdown.

Line 5: This is an example to show different levels of headings in RMarkdown. Line 6: This is an example to show different levels of headings in RMarkdown.

Question: What happens to the size of text when you increase the number of hashtags (#)?

Answer: Increasing numbers of hashtags decreases the font size.

In the example above, the higher the number of hashtags, the lower the heading level (smaller font).

Question: How many hashtags would you use if you wanted highest level heading (largest font)?

Answer: 1

Question: In the rendered output, do Line 5 and Line 6 from above example appear in the same line?

Answer: Yes

Task: Go to the metadata for this assignment and change the output from `pdf_document` to `html_document`.

Question: Did your Line 5 and Line 6 appear in different lines?

Answer: No

Sometimes the way the program renders output differs between formats. You can leave two spaces at the end of the line for a line break.

Task: Change the output back to `pdf_document`.

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2.2 Italics

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You may sometimes want to italicize text. RMarkdown allows you to do this easily. To make your text italics, you can enclose it within single asterisk `*[text]*`.

Line 1: This is an example to show how to make text italics.

Task: Rewrite Line 1 below and only italicize the word “italics.” Leave two spaces at the end of sentence for a line break.

Answer: Line 1: This is an example to show how to make text *italics*.

You may be wondering if there is another way to make the text italics, and in fact, there is. You can also enclose your text within single underscore `_[text]_`.

Task: In the paragraph below, make “Arnav,” “Biological Data Wrangling and Visualization” and “RMarkdown” italics in the following paragraph using both the methods (asterisks and underscore) mentioned above

at least once. My name is *Arnav Pandey*. I am currently enrolled in *Biological Data Wrangling and Visualization* (BIO 457/657). This is a task to use *RMarkdown* to make a text output italics.

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2.3 Bold

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Sometimes you may want to bold your text to emphasize its importance. RMarkdown supports that as well.

Task: Read the question below, click Knit, and return to answer the question below. Question: Which of the following lines gives you bold text?

Line 1. This is an example to show how to make text bold.

Line 2: This is an example to show how to make text bold.

Answer: Line 2

You can use two asterisks before and after the text to make it bold (****[text]****).

There is also another way to make a text bold by enclosing your text within two underscores(**__[text]__**).

Don't believe me? Knit to see if the line below is bold.

Line 3: This is an example to show how to make text bold.

Like you practiced with italics, you can make any word or phrase within a sentence bold. See example of it below when you hit Knit.

Line 5: This is an example to show how to make text **bold**.

Line 4: This is an **example** to show how to make text bold.

Task: In the paragraph below, make “Arnav,” “Biological Data Wrangling and Visualization” and “RMarkdown” bold in the following paragraph using both the methods (asterisks and underscores) mentioned above at least once.

My name is **Arnav** Pandey. I am currently enrolled in **Biological Data Wrangling and Visualization**(BIO 457/657). This is a task to use **RMarkdown** to make a text output bold.

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2.4 Bold and Italics

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Is there a way to make a text both bold and italics at the same time? Indeed, there is! You can enclose your text within three asterisks (*****[text]*****). or underscores (**___[text]___**).

Task: In the paragraph below, make “Arnav,” “Biological Data Wrangling and Visualization” and “RMarkdown” bold italics using both the methods mentioned above at least once.
My name is ***Arnav*** Pandey. I am currently enrolled in ***Biological Data Wrangling and Visualization*** (BIO 457/657). This is a task to use ***RMarkdown*** to make a text output bold.

In sections 2.2, 2.3, and 2.4, you learnt how to make text italics, bold, or bold+italics by enclosing the text within one, two, and three asterisks or underscores, respectively. You might be tempted to start enclosing a text by underscore and end it using asterisk. For example:

```
*italics_  
__bold**  
***bold italics__
```

Above lines will not give your desired output, so you will have to be consistent with how you enclose your texts. Use the same symbols at both ends!

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2.5 Bulleted list

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In RMarkdown, you can list your items in the RMarkdown as an unordered (bulleted) list.

To do that, you can follow these steps. Step 1: leave an empty line before list. Step 2: Use ‘+’, ‘-’, or ‘*’ followed by single space before you write your list item.

Step 3: To add another item, leave two spaces at the end of the previous item and go to a new line below and repeat Step 2.

Task: Knit to see how the items below appear in a list in the output.

- Apple
- Banana
- Raspberry

You can also nest subitems within an item. If you want a nested list as such, just indent the subitem that you want within an item with two spaces. Below is an example of it.

Task: Hit Knit to look at the output of the lines below:

- Apple
- Banana
- Berries
 - Blueberries
 - Strawberries
 - Raspberries
 - Blackberries
 - Cranberries
- Papaya

Task: Copy the above list below, add an item “Melon” and nest different types of melons as its subitems. Hit Knit to see your output.

- Apple
- Banana
- Berries
 - Blueberries
 - Strawberries
 - Raspberries
 - Blackberries
 - Cranberries
- Papaya
- Melon
 - Watermelon
 - Cantaloupe
 - Honeydew
 - Canary Melon

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2.6 Numbered List

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Getting a numbered list is even easier compared to unordered or bulleted list. To generate a numbered list, you can follow these steps. Step 1: Leave an empty line before list. Step 2: Use uppercase/lowercase English alphabets, Arabic numerals, uppercase or lowercase roman numerals followed by space before you write your list item. Be consistent how you want to list. Step 3: To add another item, leave two spaces at the end of the previous item and go to a new line below and repeat Step 2.

- I. Apple
- II. Banana
- III. Raspberry

You can also nest subitems within an item. If you want a nested list as such, just indent the subitem that you want within an item with three spaces. You can use different order format for your subitems than your primary item as well. Below is an example of it. **Please be careful with spaces!**

- 1. Apple
- 2. Banana
- 3. Berries
 - 1. Blueberries
 - 2. Strawberries
 - 3. Raspberries
 - 4. Blackberries
 - 5. Cranberries
- 4. Papaya

Note: If you want to use Roman Numerals, you may have to leave an empty line between items on list.

Section 4: Add Visualizations and Tables to Your Report

Goal

In this section, you will create graphs and a summary table using the mtcars data set. You will also practice describing patterns that you observe in the output.

Before beginning, make sure you can access the mtcars dataset. Run the line below to preview the first few rows.

```
head(mtcars)
```

Also make sure you have Tidyverse installed! Run the line of code below.

```
library(tidyverse)
```

Part A: Create a Scatterplot

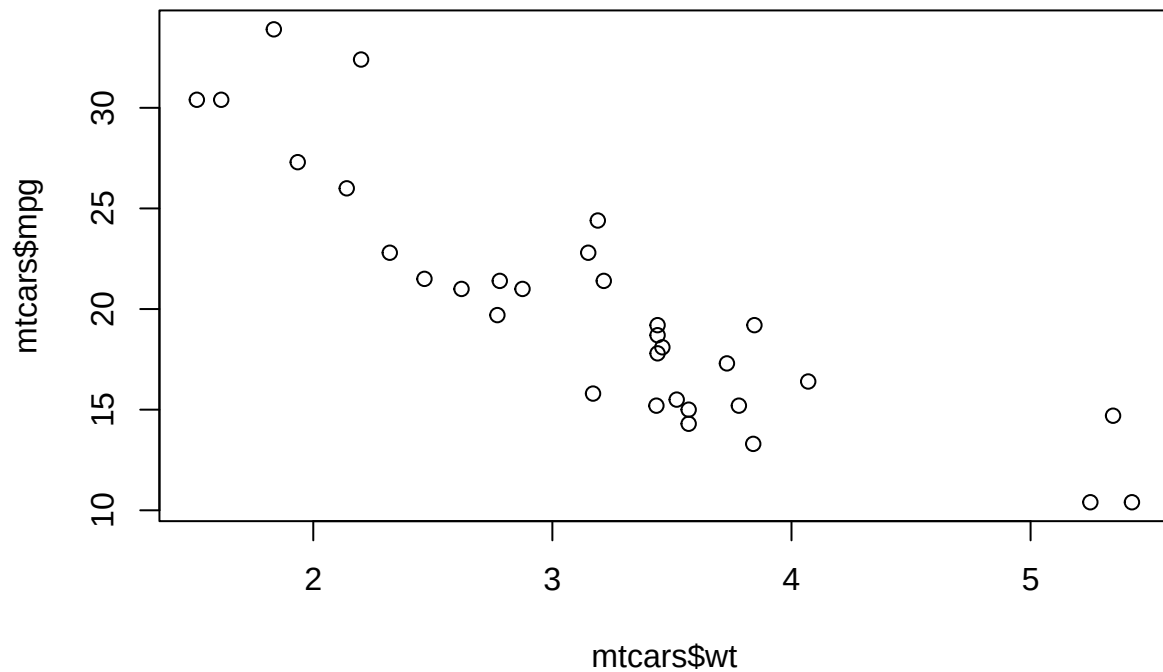
The variables wt and mpg in the mtcars data set can help us explore the relationship between car weight and fuel efficiency.

Task

Create a scatter plot with wt on the x-axis and mpg on the y-axis.

Hint: use the plot() function

```
plot(mtcars$wt, mtcars$mpg)
```

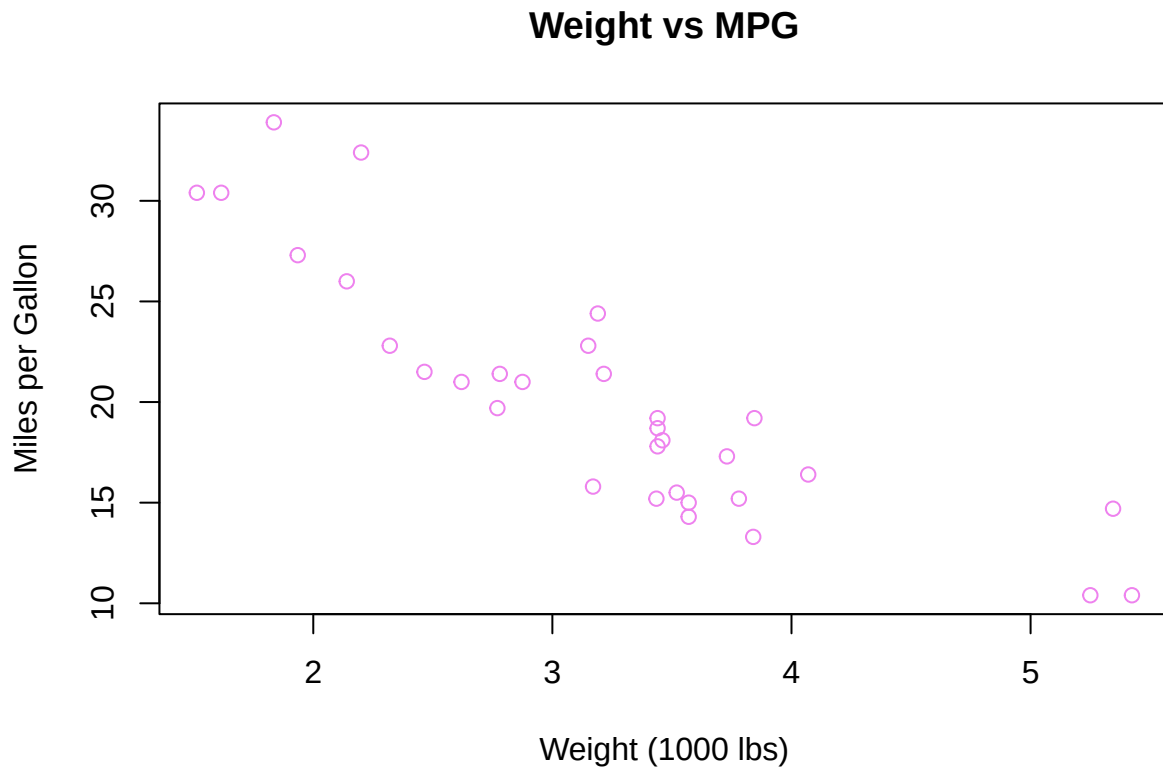


Task

Modify your scatterplot so that it includes: a title, an x-axis label, a y-axis label, a point color of your choice

Hint: Think back to previous assignments(`main=`, `xlab=`, `ylab=`, and `col=`)

```
plot(mtcars$wt, mtcars$mpg,  
     main = "Weight vs MPG",  
     xlab = "Weight (1000 lbs)",  
     ylab = "Miles per Gallon",  
     col = "violet")
```



Question

What does each point in this scatterplot represent?

Your answer: Each point represents one car in the dataset and shows that car's weight and fuel efficiency (mpg).

Question

What relationship do you observe between weight and mpg?

Your answer: As weight increases, mpg decreases. Heavier cars tend to have lower fuel efficiency.

Question

Does the graph suggest a positive relationship, a negative relationship, or no relationship? Explain your answer.

Your answer: The graph shows a negative relationship because the points trend downward as weight increases, meaning mpg decreases as weight increases.

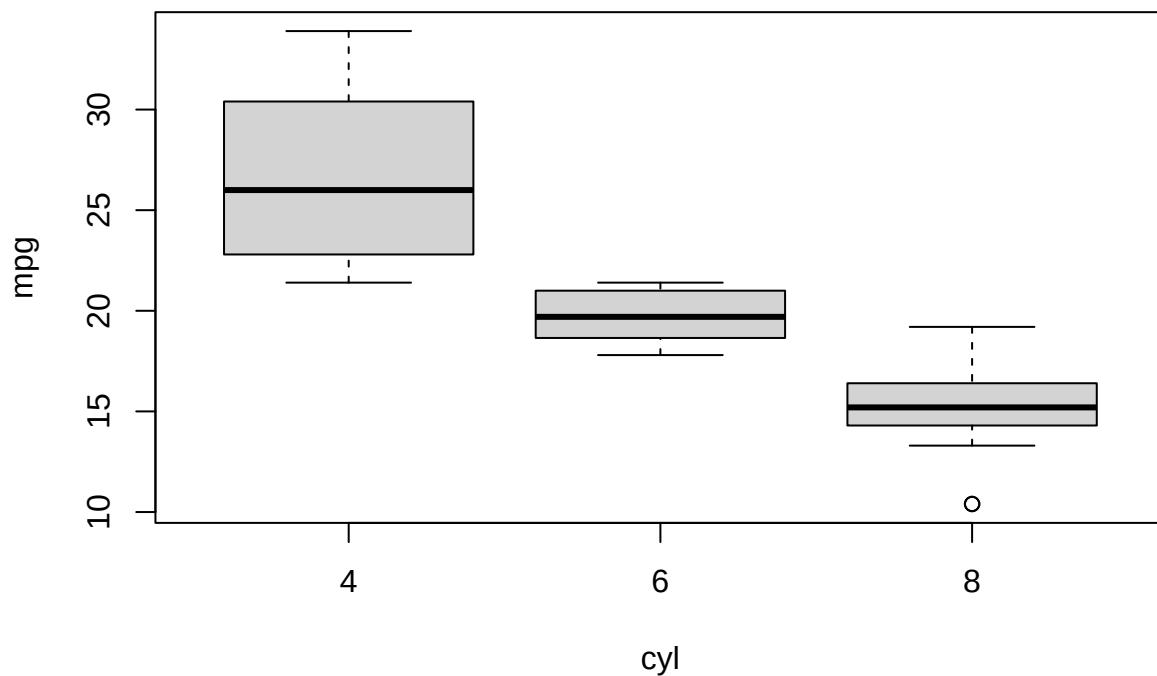
Part B: Create a Boxplot

The variable `cyl` shows the number of cylinders in each car. A boxplot can help compare fuel efficiency across cylinder groups.

Task

Create a boxplot of `mpg` by `cyl` using the `boxplot` function

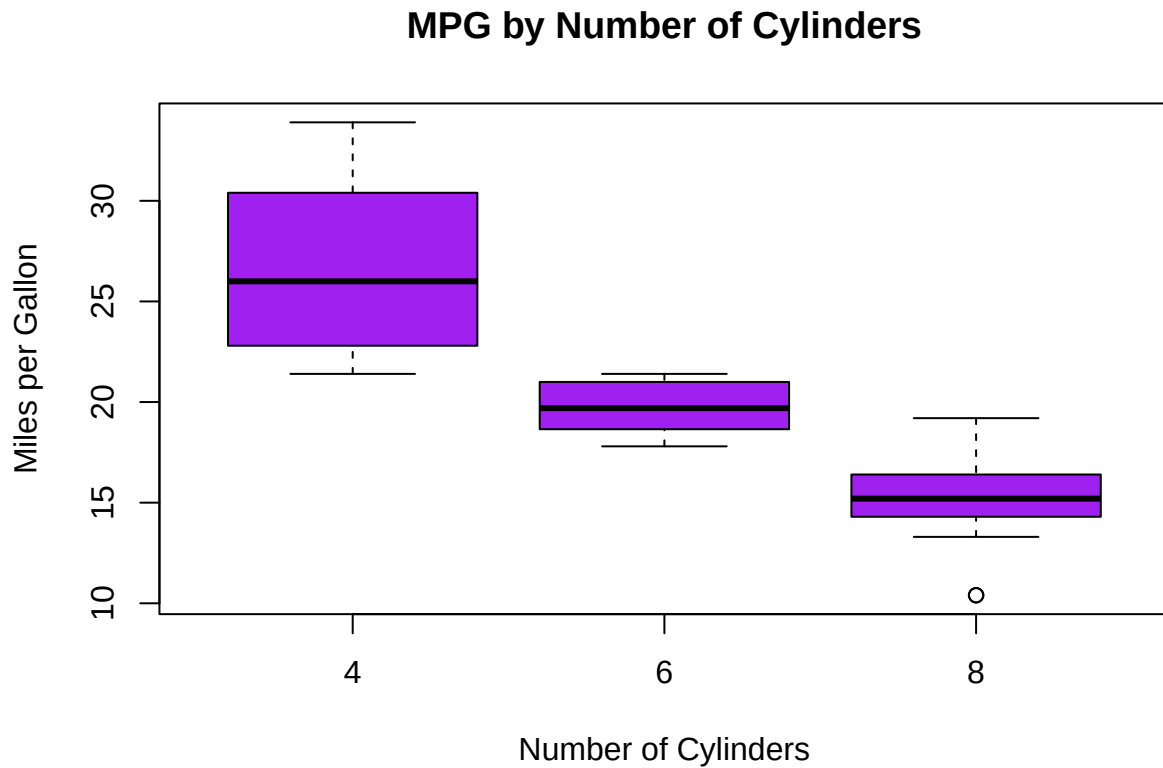
```
boxplot(mpg ~ cyl, data = mtcars)
```



Task

Modify your boxplot so that it includes: a title, an x-axis label, a y-axis label, and a fill color of your choice

```
boxplot(mpg ~ cyl, data = mtcars,  
        main = "MPG by Number of Cylinders",  
        xlab = "Number of Cylinders",  
        ylab = "Miles per Gallon",  
        col = "purple")
```



Question

What does each box in this plot represent?

Your answer: Each box represents the distribution of mpg values for a specific cylinder group.

Question

Which cylinder group appears to have the highest fuel efficiency?

Your answer: The 4-cylinder group has the highest fuel efficiency

Question

Which cylinder group appears to have the lowest fuel efficiency?

Your answer: The 8-cylinder group has the lowest fuel efficiency.

Question

Based on the boxplot, how does fuel efficiency seem to change as the number of cylinders increases?

Your answer: Fuel efficiency decreases as the number of cylinders increases. Cars with more cylinders tend to have lower mpg.

Part C: Create a Summary Table

You will now create a table showing the average miles per gallon for each cylinder group.

Task

Use dplyr to calculate the average mpg for each value of cyl. Rename the data avg_mpg_by_cyl

Hint: you need to group the data by cyl then summarize the mean of mpg.

Use the space below to write your code:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.6.0
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
avg_mpg_by_cyl <- mtcars %>%
  group_by(cyl) %>%
  summarise(avg_mpg = mean(mpg))
```

Task

Display your results as a clean table.

Use the space below to write your code:

```
avg_mpg_by_cyl
```

```
## # A tibble: 3 x 2
##   cyl avg_mpg
##   <dbl> <dbl>
## 1     4  26.7
## 2     6  19.7
## 3     8  15.1
```

Question

What does this table show?

Your answer: The table shows the average miles per gallon for each cylinder group, summarizing fuel efficiency across the groups.

Question

Which cylinder group has the highest average mpg?

Your answer: The 4-cylinder group has the highest average mpg.

Question

Which cylinder group has the lowest average mpg?

Your answer: The 8-cylinder group has the lowest average mpg.

Part D: Compare the Visuals

Now that you have created both graphs and the table, compare the information they provide.

Question

How does the scatterplot help you understand the relationship between weight and fuel efficiency?

Your answer: The scatterplot shows individual data points and helps visualize the relationship between weight and mpg, showing that heavier cars have lower fuel efficiency.

Question

How does the boxplot help you compare different groups of cars?

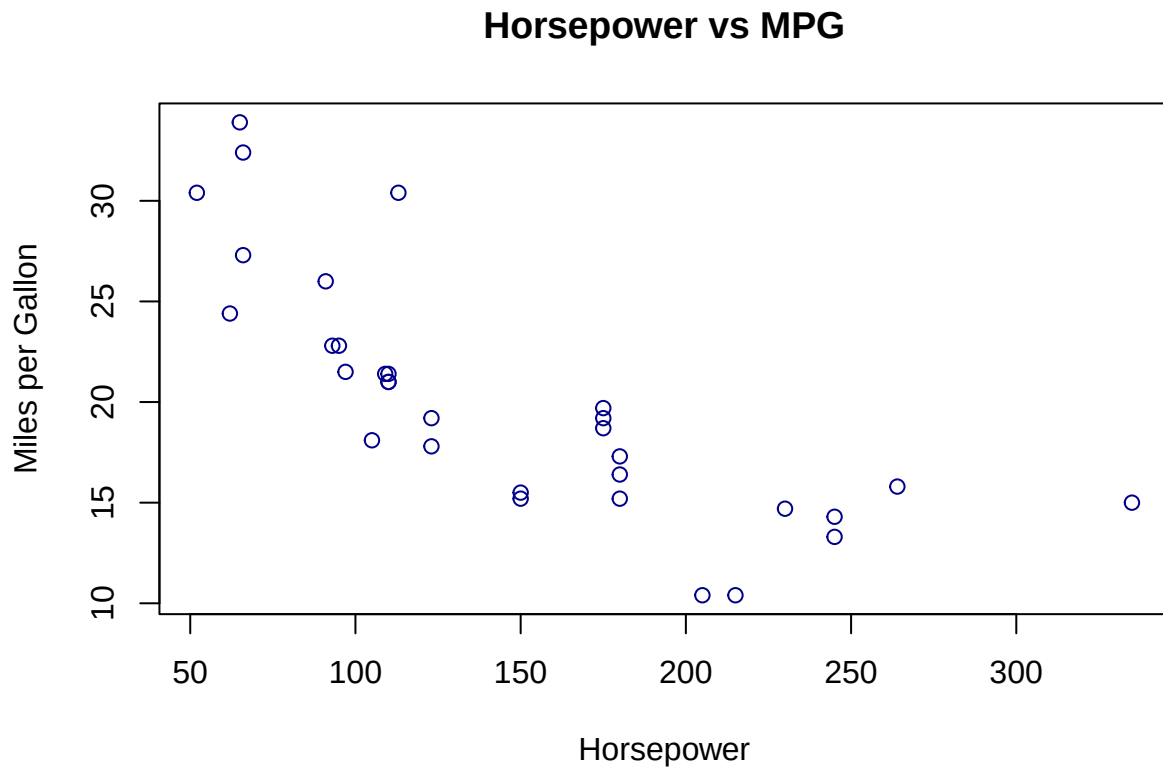
Your answer: The boxplot groups cars by cylinder count and shows the spread of mpg within each group, making it easier to compare fuel efficiency across different groups.

Part E: Now you try

1. Use a different variable from mtcars to make an additional scatter plot.
2. Change the color of your scatterplot.

Write your code below:

```
plot(mtcars$hp, mtcars$mpg,  
     main = "Horsepower vs MPG",  
     xlab = "Horsepower",  
     ylab = "Miles per Gallon",  
     col = "darkblue")
```



Question

What did you change, and how did it affect the output?

Your answer: I changed the x-variable from weight to horsepower and changed the color of the scatterplot. This allowed me to explore a different relationship in the data and improved the visual appearance of the graph.